



BALL STATE
UNIVERSITY

Module 3 Lecture - Probability Topics

Introduction to Statistical Methods

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Table of Contents

1	Overview and Introduction	2
1.1	Learning Objectives	2
1.2	Introduction	2
2	Terminology	3
2.1	Introduction	3
2.2	Rapid Fire Vocabulary	3
2.3	Calculating Probability	5
2.4	OR Events	6
2.5	AND Events	7
2.6	Conditionals and Complements	8
3	Independent and Mutually Exclusive Events	9
3.1	Introduction	9
3.2	Independence	9
3.2.1	Connection to Sampling Process	9
3.3	Mutually Exclusivity	10
4	Contingency Tables	11
4.1	Introduction	11
4.2	Example of a Contingency Table	11
5	Tree and Venn Diagrams	11
5.1	Introduction	11
5.2	Tree Diagrams	12
5.3	Venn Diagrams	13
6	Conclusion	13
6.1	Recap	13
	Key Terms	13

1 Overview and Introduction

1.1 Learning Objectives

- Understand and use the terminology of probability.
- Determine whether two events are mutually exclusive and whether two events are independent.
- Calculate probabilities using the Addition Rules and Multiplication Rules.
- Construct and interpret Contingency Tables.
- Construct and interpret Venn and Tree Diagrams.
- Begin to appreciate how probability connects back to the broader idea of statistics
- Understand the emphasis of chance and probability, not certainty, in statistics

1.2 Introduction

- Most decisions we make are a (hopefully educated) _____.
 - E.g., Should I wear my boots if it might rain today?, What are the chances that my class will respond well to this activity?
 - Chance and probability is _____, and, if we introspect closely, it plays a role in pretty much everything we do

 Discuss: Try coming up with more examples of weighing probability in your own life

- **Probability** is a core part of _____, and our inferential results are often phrases in recognizing that level of uncertainty
 - This is why you never really hear good scientists say we have definitively “_____” something - there’s always room left for uncertainty
 - In science, we are trying to demonstrate something beyond a _____ doubt
- There is a lot of _____ (and math) surrounding probability
 - You could take an entire semester long class just talking about probability and the _____/math of it!

- We'll try to stay centered on exploring it as relevant to _____, though this module may initially feel very removed from statistics

! Important

Even though we won't be 'in the weeds' as much with probability calculations later on, it will remain an important background subject!

2 Terminology

2.1 Introduction

- Like with the last module, _____ makes perfect
 - Please use the various exercises in the book and visit GA and/or professor office hours!
 - This is also a great time to embrace flashcards, if that historically works well for you

2.2 Rapid Fire Vocabulary

- In the context of probability, an **experiment** is some _____ controlled and monitored action - similar to how we use the term in research design
 - If the experiment is not **pre-determined**, that is, it's outcome is not _____, then there is some amount of **chance**
 - So if we call something a **chance experiment**, its some controlled action that we are not _____ of the outcome
 - Classic example is a coin flip

? Example of how probability are intuitive (to an extent) - what is the chance that I flip a coin and it lands on 'tails'

- ☒ A) 25%
- ☐ B) 75%
- ☐ C) 50%
- ☐ D) 0%

Explanation:

- The final _____ of the experiment is called an **outcome**
 - The possibilities of _____ the different outcomes is called the **sample space** (notation = S)
 - Sample space is _____ through **list notation**, or in a tree or venn diagram; more example of these later
 - Coin flip with possible heads (H)/tails (T) outcomes _____
notation example = $S = \{H, T\}$
 - 6-sided dice roll in list notation: $S = \{1, 2, 3, 4, 5, 6\}$
- An **event** is some _____ of outcome(s)
 - Important to know that it could just be _____ outcome or more than one described outcome
 - _____ will be defined as upper case letters like A, B, C , etc.
 - Example: Event A would be a six-sided die landing on a number 1 through 3
- The **probability** of a certain _____ occurring is described as the **long-term relative frequency** of said event

📢 Discuss: Review: from module 2, try to describe in your own words what relative frequency is?

- More on long-term relative frequency
 - What that means is that the probability describes how _____ a certain event would occur if the experiment were to be completed an infinite number of times; this hinges on the **law of large numbers**
 - The _____ of a certain event happening will be written as $P(x)$, where x is whatever event is being looked at
 - * $P(x)$ will always be ≥ 0 and ≤ 1
 - When $P(x)$ of all possible _____ are equal to one another, we would say that this experiment is **equally-likely/fair**

 Discuss: Review: from module 2, we've talked about the law of large numbers before. Try explaining in your own words how it relates to sample statistics and population parameters

- Example of coin toss, where outcome H is the likelihood of flipping with an outcome of head on a single toss, and outcome T is the likelihood of tails
 - If $P(H) = 0.50$, i.e., a 50% chance to land on heads
 - If $P(T) = 0.50$, i.e., a 50% chance to land on tails
 - If $P(H) = P(T)$, i.e., equally likely experiment

 Discuss: Try writing out the same set of bullets that I just did, but with a six-sided equally likely die instead. We will calculate it out in a second, but try to just do this intuitively now.

2.3 Calculating Probability

- In the scenario that we aim to calculate the probability of a single _____ in an equally likely _____ space:

- Steps to :
 1. Determine all outcomes from the experiment and list them out, counting the number of possible outcomes
 2. a specific event
 3. Assess how outcomes match that event
 4. the number of outcomes that match the event by the total number of outcomes; $P(x) = Matching/Total$
- Do a coin toss of a dime and nickel
 1. 4 outcomes in the sample space: $\{HH, TH, HT, TT\}$ where
 - H heads on a coin
 - T tails on a coin
 - In writing this would be: both coins land on heads, dime lands tail and nickel lands heads, dime lands heads and nickel lands tails, both coins land on tails
 1. Event A of getting at only one heads; could also be written as $A : n_H = 1$
 2. Two outcomes in the sample space match this criteria: $/TH, HT/$
 3. Thus the probability of event A is calculated as $P(A) = 2/4 = 0.50$

 Discuss: Trying doing the same full calculation process now, but now for a 6-sided equally likely die.

- Practically, many events are not fair / equal - coin flips are affected by the person doing them, air resistance, etc.
 - This will be called unfair/biased

2.4 OR Events

- An OR event is when an event is written in such a manner that an outcome could event A or B
 - This would be written as $A \cup B$ (that symbol is called a cup)
 - Applying the above, we would have a step process of determining the outcome match A OR B and would give the probability as $P(A \cup B)$

- Practical example: What is the chance that a student has a GPA above 3.00 OR an SAT score above 1400.
- To list the possible outcomes, we find the **set union** of the two sets of outcomes that match the _____ events, example:
 - If A matches $\{1, 2\}$ and
 - If B matches $\{2, 3, 4\}$ then
 - $A \cup B$ matches $\{1, 2, 3, 4\}$
 - Then we proceed by using $A \cup B$ matches in step 4 of **Calculating Probability**

2.5 AND Events

- An **AND event** is when an event is written in a manner that an outcome must match event A and B
 - This would be written as $A \cap B$ (this symbol is a cap, like a hat)
 - Thus probability would be written as $P(A \cap B)$
 - Practical example: What is the chance that a student has a GPA above 3.00 AND an SAT score above 1400.
- To list the possible outcomes, we find the **set intersection** of the two sets of outcomes that match the _____ events, example:
 - If A matches $\{1, 2, 3\}$ and
 - If B matches $\{2, 3, 4\}$ then
 - $A \cap B$ matches $\{2, 3\}$
 - Then we proceed by using $A \cap B$ matches in step 4 of **Calculating Probability**

! Important

A mnemonic to remember caps and cups: cAps are for Ands

 Discuss: Try writing out (just in words, not notation) an example of an OR and AND event relevant to your field of interest, like I did with GPA and SAT earlier

2.6 Conditionals and Complements

- **Complements** take the _____, or everything but the matching outcomes of an event
 - This is written at the _____ of an outcome, e.g., A' is the complement of A
 - Thus, probability of a conditional would be notated as $P(A')$
 - In such a scenario, instead of finding all the outcomes that DO match the event, we would list all the outcomes that _____ match the event in step 3 of **Calculating Probability**
 - Example: What is the probability that a patient does NOT have an IQ below 85?
- **Conditionals** are the probability a certain event occurs _____ that another event has already happened
 - It is written as $P(A|B)$, which translates to probability of A given B (the vertical bar is sometimes called a _____)
 - Example: If a person is determined to dislike pizza, what is the probability they will also dislike pasta?
 - This is calculated via the following formula, assuming $P(B) \neq 0$:

$$P(A|B) = \frac{A \cap B}{B}$$

- Once again, try working through the problems and solutions in your book for all of these scenarios!

 **Discuss:** Consider the following prompt: 'I am calculating the probability that a person is in the 90th percentile on reading and math abilities, given that they are established to be above the 80th percentile on writing' Which of the event types that we've discuss does this prompt use: AND, OR, Compliment, Conditional

3 Independent and Mutually Exclusive Events

3.1 Introduction

- We are often interested in how two possible events are _____ related or unrelated
 - We can investigate both the independence and mutually exclusivity of events to determine this
 - These are both _____ descriptions

3.2 Independence

- Two events are considered to be **independent** if they are unrelated/do not _____ one another
 - On the contrary, if they are somehow _____ to one another, they are said to be **dependent**
- The following are the _____ rules to show independence:
 - $P(A|B) = P(A)$
 - $P(B|A) = P(B)$
 - $P(A \cap B) = P(A)P(B)$ (The **Multiplication Rule**)
 - Only one of these conditions needs to be met for establishing independence
- It is safest to _____ dependence of events until independence has been established
 - Sort of counter-intuitive, but useful to our assumption for _____ tests, more on that later

3.2.1 Connection to Sampling Process

- Recall that in a *random* sampling scenario, there is a theoretically _____ chance for each member of the population of interest to be selected
 - Effectively, _____ sampling is sampling that is done in an *equally likely* manner
 - Example: If I am sampling US military veterans (let's say $n = 1000$), then a truly random sampling will give each of those people (outcomes) a probability of $1/1000$ to be chosen

- Sampling can be done one of two ways, that impact whether sampling decisions were made _____ of one another
 - **Sampling with replacement**: means the same individual could be chosen _____ times - this means the odds of being selected as a member of the population remain the same (e.g., selection of each individual is _____)
 - **Sampling without replacement**: means it is done in a manner that a person _____ be chosen multiple times, thus reducing the outcomes by one and changing the odds for each subsequent selection (e.g., selection of each individual is _____)
 - Marble example

! Important

Thus, our sampling is not truly 'random' unless we do it with replacement!

📢 Discuss: Consider the following prompt 'I simple random sampled 50 individuals from a larger population without replacement, does this result in a sample that should be representative of the broader population?'

3.3 Mutually Exclusivity

- Events are **mutually exclusive** when they cannot occur at the same time and follow the **Addition Rule**:

$$P(A \cap B) = 0.00$$

- If unsure, assume _____ until this rule is met

4 Contingency Tables

4.1 Introduction

- The word “contingent” means to depend on, so **contingency tables** offer a nice view for displaying probabilities in such a way to dependence

! Important

Like with descriptive statistics, using graphical and tabular displays can make it much easier to navigate complex information, make sure you use them!

4.2 Example of a Contingency Table

	Speeding violation in the last year	No speeding violation in the last year	Total
Uses cell phone while driving	25	280	305
Does not use cell phone while driving	45	405	450
Total	70	685	755

Table 3.3

5 Tree and Venn Diagrams

5.1 Introduction

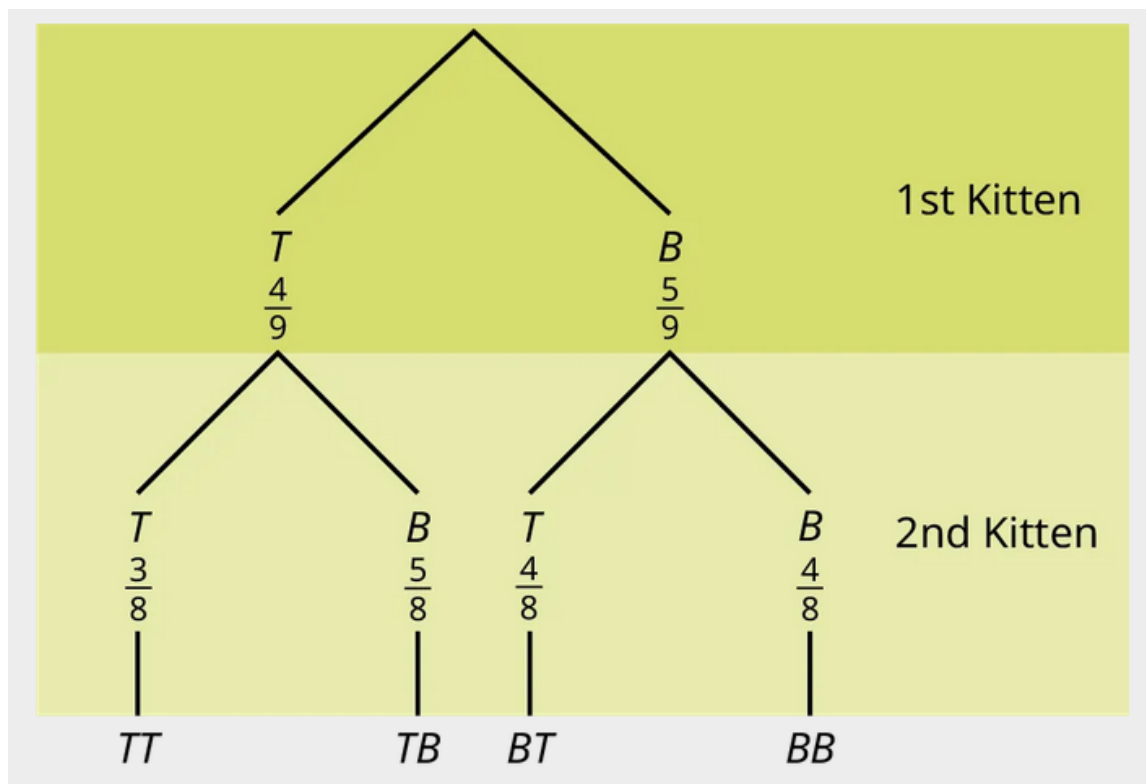
- Just like with how descriptive statistics have several ways to graphically present, so do probabilities!

- Contingency tables are a great start, but we can add to them with _____ diagrams and _____ diagrams
- For this section, I'll be a bit more brief only because these representations are not super _____ in most applied statistics

! Important

In scenarios with large sample spaces, the tree and venn diagrams may offer more practical representations, but realistically list notation works fine for most scenarios

5.2 Tree Diagrams



5.3 Venn Diagrams

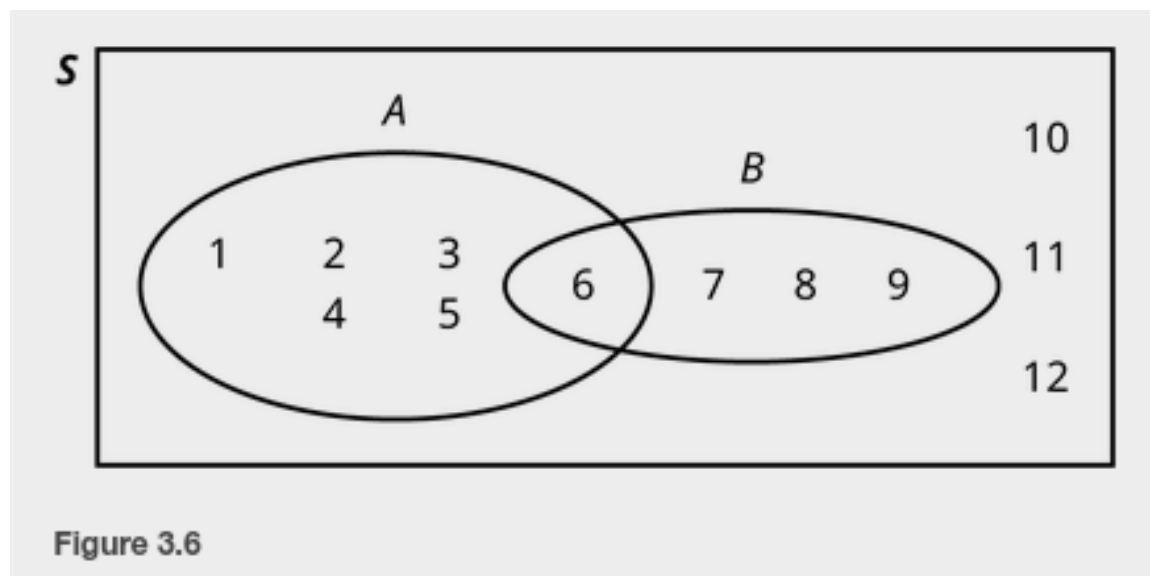


Figure 3.6

6 Conclusion

6.1 Recap

- Probability is a tough subject, but it is critical as a foundation topic prior to understanding inferential statistics!
- Understand that when we discuss statistical results going forward, it will always be in through the lens of probabilities and likelihood of outcomes.
- While “real” statistics won’t often involve intensive hand calculation of probabilities, you should still have a foundation appreciation of how to do it - just like we did with standard deviation.

Key Terms

A

Addition Rule A formulation for determining that two or more events are mutually exclusive 10

AND event An event written in such a way that outcomes must match both events 7

C

chance A general term to encapsulate the idea that there are varying likelihoods of certain outcomes from an experiment 3

chance experiment Some experiment in which an outcome is not pre-determined or certain 3

Compliments Taking the probability of all outcomes opposite of a stated event 8

Conditionals Taking the probability that a certain event occurs given another event has already happened 8

contingency tables A view of outcomes/events that lists their relative frequencies 11

D

dependent If two events are somehow related to or affected by one another 9

E

equally-likely/fair When all possible outcomes from an experiment are equal to one another 5

event Some pre-defined criteria that we are looking to see if our outcomes meet 4

experiment Some pre-defined trial that has a set number of possible outcomes 3

I

independent If two events are unrelated to one another 9

L

list notation A way to represent sample space in a comma-separated list 4

long-term relative frequency The rough proportion of times that an event would occur if the experiment was run an infinite number of times 4

M

Multiplication Rule A formulation for establishing independence between two events 9

mutually exclusive When events cannot occur at the same time 10

O

OR event An event written in such a way that outcomes may match one or more of the events 6

outcome The result of an experiment 4

P

pre-determined When an experiment has an outcome that is certain to happen 3

Probability A measure of the likelihood of a certain event occurring 2

S

sample space The possibilities of all of the different outcomes from an experiment

4

Sampling with replacement A process for sampling in which the same result can be chosen multiple times 10

Sampling without replacement A process for sampling in which the same result cannot be chosen multiple times 10

set intersection A step in calculating AND events, used to combine outcomes only shared between the two events 7

set union A step in calculating OR events, used to combine all outcomes from all events 7

U

unfair/biased When an experiment has unequal probability of outcomes 6

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in your book and your own judgment to make sure you have all relevant terms